Ergan Shang

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Education

Carnegie Mellon University *Pittsburgh, PA, USA*
Ph.D. in Statistics, Advisors: Kathryn Roeder, Yuting Wei, Weijing Tang *08/2023 - (exp.) 05/2028*

University of Science and Technology of China *Hefei, Anhui, China*
B.S. in Mathematics, GPA: 4.07/4.30 (Top 2%) *08/2019 - 06/2023*

Work Experience

- **Research Scientist Intern, Ads Foundation Model and Ranking Engineering, Meta Platforms, Inc.** *05/2026 - 08/2026*
  Under supervision of Flavio Sales Truzzi in **Training Efficiency Team**. **Submitted to NeurIPS 2026** [Paper]
  - Proposed a **Forward-Forward Early Backward** training algorithm to decouple forward/backward architectures for parameter updates, validating correctness on **NanoChat transformer** models via GPU kernel/operator trace analysis.
  - Conducted **low-fidelity proxy evaluations** to screen training-efficiency proposals, validating that the proxy introduced only **0.0029% model-performance bias**.
  - Applied on Meta's **production-scale Dynamic Product Ads (DPA) CTR model** for high-traffic ads ranking, yielding a **7.38% QPS(query per second)** gain while keeping **model performance** neutral within **1.38%** measured by normalized entropy (NE).

Research Experience

- **Comparative-signal-based GRPO fine-tuning for mathematical reasoning** *12/2025 - present*
  Under supervision of Linjun Zhang.
  - Proposed an EIF(*efficient influence function*)-corrected **GRPO objective** using pairwise LLM comparison signals, where auxiliary solutions are judged by a third LLM to construct preference-based influence scores for policy optimization.
  - Implemented the full simulation and mathematical reasoning pipeline, **fine-tuning Qwen-0.5B** on **GPQA and GSM8K** with **GPT-4o-mini** as the auxiliary LLM for pairwise solution evaluation, achieving a **10% improvement in pass@k** over the naive baseline.
- **Co-author, ASI-Bench [Hugging Face #4 Paper of the day, Aim for Nature] [Web]** *Apex Intelligence | 07/2026 - 09/2026*
  - Designed the task to evaluate agents' reasoning on high-dimensional feature selection via reward-based grading; frontier agents scored only **20/100**, highlighting the challenge of scientific reasoning beyond code completion.
- **AI Agent Benchmarking for Multi-Step Scientific Reasoning** **Submitted to ICLR 2027[Paper]** *07/2026 - 09/2026*
  - Contributed a life-sciences task to **Long-Horizon Terminal-Bench (LHTB)**, proposing a project-level AI-agent benchmark of **13/100** scores with parameterized synthetic data, hidden verifiers, and reward-based scorers.
- **A diffusion-based debiasing framework for prediction at single-cell resolution** *11/2024 - 09/2025*
  Under supervision of Kathryn Roeder. **Accepted by Proceedings of the National Academy of Sciences of the US(PNAS)**: [Paper]
  - Developed the full pipeline of an accelerated **diffusion model** with **VAE** to generate single-cell gene profiles under unseen perturbations, producing ∼**30000 samples in minutes versus** ∼**100** in standard biological experiments.
  - Incorporated statistical debiasing techniques, achieving **85% mean estimation coverage** compared with ∼80% or lower for competitor methods, substantially reducing time and cost.
- **Transformer-Based Transfer Learning for High-Dimensional Response Prediction** *12/2025 - 05/2026*
  Under supervision of Kathryn Roeder and Yuting Wei.
  - Developed **SCARF**, a deep generative transfer-learning framework that combines **Transformer-based representation refinement**, **optimal transport** alignment, and **VAE** decoding to transfer distributional shifts and predict noisy high-dimensional responses under unseen experimental contexts.
  - Applied SCARF to multiple real benchmark datasets, achieving an average **PCC**(Pearson Centered Coefficient) of **0.95** and outperforming multiple classic baselines, while reducing GPU memory usage by approximately **50%**.
- **Inference for Balance Theory in Time-Varying Signed Network** *11/2023 - 12/2025*
  Under supervision of Weijing Tang. **Submitted to Journal of Machine Learning Research (JMLR)**: [Paper]
  - Developed a novel **nonparametric model** integrating dynamic and signed networks, implementing a **PyTorch-based kernel smoothing method** to estimate network balance and tune model parameters.
  - Achieved a **7x precision improvement** in simulations and applied the model to international politics data to **detect anomalous relationships** (posters: **S**tatistical **N**etrowk **A**nd **B**eyond Tokyo **06/2025**; ASA Pittsburgh **04/2025**).
- **XConTest: Convergence Analysis of Perturbation Effects for Gene Function Similarity** *11/2023 - 12/2025*
  Under supervision of Kathryn Roeder and Tianyu Zhang. **Submitted to Science Advances**: [Paper], [Software]
  - Developed an **application-driven** framework for gene function similarity, combining a two-step LASSO pipeline with high-dimensional mean comparison, projection, and cross-fitting to identify relevant perturbations and interactions.
  - Achieved **over 90% accuracy** in simulations compared with competitor methods and applied the framework to real datasets to detect pathological gene activity; gave an **Invited Talk** at 2024 Joint Statistical Meetings **08/2024**.

Coursework

- **Machine Learning:** Advanced ML Theory&Method(A), Convex Optimization(A), Statistical Methods in ML(A);
- **Programming:** Applied Statistical Software(R Language), Data structure and Database, C and Algorithm.
- **Statistics:** Intermediate Statistics(A+), Advanced Statistical Theory(A), Advanced Data Analysis, Stochastic Process, Multivariate Analysis, Regression Analysis(A+)

Awards

- Mihaela Serban Award for the **Best Poster Competition** in ASA Pittsburgh Chapter; *2025*
- **Candidate of Guomoruo Scholarship (Highest Award in Undergraduates) (Final Round, Top 2%);** *2022*
- **Meritorious Award** in Mathematical Contest In Modeling and Interdisciplinary Contest In Modeling (MCM/ICM) (Top **8**% in the participating teams **Worldwide**); *2021*

Technical Skills

- **5+ Years** of research experience in **data science, statistics, machine learning**, with expertise in **PyTorch**, generative and foundation models, **Transformers**, **U-net**, and LLM-based frameworks like **Reinforcement Learning, fine-tuning** Algorithm, PPO, GRPO.
- **5+ Years** of programming experience in **Python, R, and SQL** for data analytics, statistical modeling, and large-scale data processing.